Restorative Neurology and Neuroscience 33 (2015) 571–578
DOI 10.3233/RNN-130334
IOS Press

Electrical stimulation and testosterone enhance recovery from recurrent laryngeal nerve crush

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Abstract.
Objective: This study investigated the effects of a combinatorial treatment, consisting of a brief period of nerve electrical stimulation (ES) and systemic supraphysiologic testosterone, on functional recovery following a crush of the recurrent laryngeal nerve (RLN).
Study design: Prospective, controlled animal study.
Methods: After a crush of the left RLN, adult male Sprague-Dawley rats were divided into four treatment groups: 1) no treatment, 2) ES, 3) testosterone propionate (TP), and 4) ES + TP. Each group was subdivided into 1, 2, 3, or 4 weeks post-operative survival time points. Groups had an *n* of 4–9. Recovery of vocal fold mobility (VFM) was assessed.
Results: Brief ES of the proximal nerve alone or in combination with TP accelerated the initiation of functional recovery. TP administration by itself also produced increased VFM scores compared to controls, but there were no statistical differences between the ES-treated and TP-treated animals. Treatment with brief ES alone was sufficient to decrease the time required to recover complete VFM. Animals with complete VFM were seen in treatment groups as early as 1 week following injury; in the untreated group, this was not observed until at least 3 weeks post-injury, translating into a 66% decrease in time to complete recovery.
Conclusions: Brief ES, alone or in combination with TP, promise to be effective therapeutic interventions for promoting regeneration following RLN injury.

Keywords: Axotomy, regeneration, androgen, electrical stimulation, rat, recurrent laryngeal nerve, functional recovery, vocal fold

1. Introduction

The RLN follows a long and indirect course from where it branches off the Vagus nerve to its laryngeal entry. Its passage through the thorax and course

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through the thyroid gland make it particularly vulnerable to damage during surgery involving the neck or thorax. RLN lesions result in significant patient morbidity, causing vocal fold paralysis and incomplete glottic closure which can lead to changes in vocal pitch or a lack of voice, dysphonia, dysphagia, and in some cases aspiration (Myssiorek, 2004; Rosenthal et al., 2007). The effects of these symptoms on patient quality of life can be devastating and recovery from this type of injury is variable and dependent on the degree of the initial damage; some patients recover spontaneously within weeks while others never do. Current treatments focus on medialization of the injured vocal fold through injection of collagen but do not address the lack of mobility (Rubin et al., 2003). Intraoperative monitoring of nerve conductivity is standard procedure in operating rooms during surgeries that put the RLN at risk; however, this has only been used as a pre-emptive measure. Previous work has demonstrated that a combination treatment consisting of brief electrical stimulation (ES) and supraphysiologic levels of testosterone, a gonadal steroid, improves functional recovery from both extratemporal and intratemporal facial nerve injury in rats (Lal et al., 2008, Hetzler et al., 2008; Sharma et al., 2010b). If this combination treatment proves to be effective in a second nerve injury model, it would provide more evidence for its potential use over a spectrum of human peripheral nerve injuries.

Electrical stimulation (ES) of injured nerves is associated with enhanced regeneration and thus it would be valuable to explore the translational potential of ES to RLN injury (Gordon et al., 2008). A summary of ES-mediated neuronal effects includes neuronal excitation, upregulation of regeneration-associated gene expression, promoting neurite growth after injury, improving selective targeting of regenerating axons, providing alignment and polarity instructions to myelinating Schwann cells, and enhancing the functional recovery of target organs (Gordon et al., 2008; Wan et al., 2010). Gonadal steroid hormones have also been explored as neurotherapeutic agents and have been shown to play trophic and protective roles in conditions of injury or disease (Fargo et al., 2008). As gonadal steroids and ES may target regeneration events through different mechanisms or during different phases, combining both of these treatment strategies may be advantageous.

ES can be applied to a variety of places along the path of conduction. Target muscles innervated by the injured nerve can be stimulated or the neuron itself can be stimulated either distal or proximal to the injury site. End-organ stimulation has been clinically useful because it is a substitute for the usual neuronal input and has the effect of attenuating the extent of denervation atrophy while the injured axons are regenerating; however, it does not stimulate axonal regrowth itself (Williams, 1996). Stimulation to the injured nerve sends impulses orthodromically along the myelin sheath as well as antidromically toward the cell body to augment the internal distress signals carried by reverse axonal transport that increase expression of regeneration-associated genes such as growth-associated protein-43 (GAP-43; present at the tips of growth cones), α_1-tubulin (a structural protein within axons), brain-derived neurotrophic factor (BDNF), neuritin (enhances neurite extention *in vitro*), and pituitary adenylate cyclase-activating peptide (PACAP) (Sharma et al., 2010b; Wenjin et al., 2011).

ES has been shown to be particularly effective in accelerating rate of recovery in axonotmetic injuries such as facial and sciatic nerve crushes (Lal et al., 2008; Sharma, 2009). More severe neurotmetic injuries such as transection and prolonged compression injuries which can result in muscle atrophy and a marked decrease in the ability to successfully reinnervate target musculature can be at least partially ameliorated by treatment with ES. Clinically, a one-time, hour in duration, treatment with ES has been shown to be effective in accelerating axon regeneration and muscle reinnervation in patients with carpal tunnel syndrome (Gordon et al., 2009; 2010). In studies using animal models, facial or sciatic nerve transection injury followed by immediate repair and treatment with ES and showed improvement in motor axon regeneration (Al-Majed et al 2000) and functional recovery (Hadlock et al., 2010). More severe injuries in which regenerating axons had to cross a gap, usually through a conduit or graft, are still amenable to treatment with ES which was effective in improving the specificity and length of time of nerve regeneration (Huang et al., 2010; Haastert-Talini et al., 2011; Singh et al., 2012).

ES treatment affects "early" events during regenerations: it reduces the initial delay before sprout formation begins and leads to rapid but short-term upregulation of regeneration-associated genes which in turn improves functional recovery (Sharma et al., 2009; 2010b). Thus, the positive effects of ES on recovery have often been observed soon after treatment.

Testosterone, on the other hand, has been shown to accelerate the axonal regeneration rate through upregulation of genes involved in producing the structural building blocks of axons (including the βII-tubulin) and leads to delayed but long-term upregulation of other regeneration-associated genes (Sharma et al., 2009; Sharma et al., 2010b; Jones et al., 1997). We have previously combined this delayed effect of testosterone, using testosterone propionate (TP) which is a long-lasting form, with the more-rapidly acting ES to yield positive recovery results in facial nerve injury models (Hetzler et al., 2008; Sharma et al., 2010a). It is additionally important to note that the RLN is similar to the facial nerve in that both cranial nerve nuclei from which the two nerves arise are abundant in androgen receptors (the nucleus ambiguus and the facial motor nucleus, respectively) (Perez & Kelley, 1996; Yu & McGinnis, 2001) and thus may be similarly sensitive to androgen treatment. However, cancers of the neck and throat are typically androgen-receptor negative (Voelter et al., 2008; Hagedorn & Nerlich, 2002) – an important factor to consider when it is often the removal of these cancers during which the RLN is injured.

It has been previously demonstrated that the natural timecourse for recovery from an RLN crush injury is up to 6 weeks (Tessema et al., 2009). The aim of the present study was two-fold: 1) to develop a model for therapeutic application of ES to an injured RLN; 2) to compare the effects of individual ES and testosterone treatments and the combination on functional recovery thereby adding generality to the efficacy of the combination treatment. Brief ES and TP were administered either alone or in combination and vocal fold mobility was assessed at different time-points in different groups of rats between 1 and 4 weeks post-operative (wpo).

2. Materials and methods

2.1. Animals & nerve injury paradigm

Adult male Sprague-Dawley rats (∼2 months old) were purchased from Harlan (Indianapolis, IN) and used for all experiments. Animals were housed under a 12hr light/dark cycle and received a standard rodent diet and water ad libitum. All animals received humane care in compliance with the U.S. National Research Council's *Guide for the Car and Use of Laboratory* Animals and all animal work performed in this study was approved by the Institutional Animal Care and Use Committee of the Edward Hines VA Hospital, Hines, IL. Three to five days prior to nerve injury, rats were anesthetized with isofluorane and castrated. For RLN crush axotomies as well as vocal fold mobility assessments, rats were anesthetized with isofluorane and breathing rate was monitored continuously during procedures.

For the crush injury, a vertical incision was made in the skin of the neck, midline strap muscles were separated and retracted, and both left and right RLNs were visualized along the trachea. The left RLN was gently freed from its connective tissue attachment to the trachea and crushed at the level of the 7th tracheal ring. Two successive 30-second crushes, on alternating sides, were done with fine jeweler's forceps to ensure a full crush. To decrease the possibility of inter-animal injury variablilty, all nerve crushes are performed by the same person. This has proven to be a reliable and reproducible method that works across species and nerves (Kujawa et al., 1989; Brown et al., 1999; Serpe et al., 2002; Foecking et al., 2012; Brown et al., 2012). After verification of the crush with direct laryngoscopy, muscle layers were reapproximated and the overlying skin closed with wound clips.

Animals were divided into four experimental groups based on the type of treatment administered: [1] no treatment/control (CTL), [2] ES only, [3] TP only, and [4] ES + TP. These groups were then subdivided into time points of 1, 2, 3, and 4 weeks post-operative. Each group contained an *n* of 4–9 (148 total animals in the study). Animals were euthanized by ketamine/xylazine overdose and intra-cardiac perfusion with saline followed by ice-cold 4% paraformaldehyde solution.

2.2. Electrical stimulation

A custom electrode constructed of two Teflon-coated wires, bared of insulation for 2–3 mm, soldered to two "male" connector pins in a connector strip (Allied Electronics) were used in our laboratory. Anesthetized animals received a one-time ES treatment during the axotomy surgery. Prior to administration of an RLN crush injury, the stimulating electrode was bent around the freed RLN at the level of the 9th tracheal ring to hold it in place proximal to the crush site. Proximal is defined as closer to the cell body of the neuron (as opposed to closer to the target musculature). The grounding electrode was sutured into the retracted strap muscles 1 cm away. See Fig. 1 for a depiction of the electrode placements. The RLN

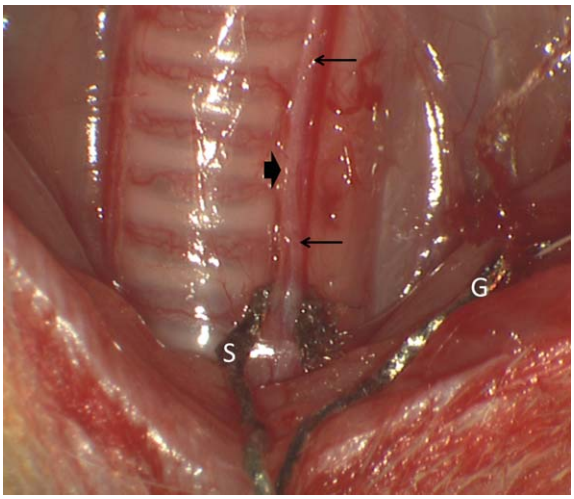


Fig. 1. Photograph of the electrode placement for nerve stimulation just prior to nerve crush. View is from anatomical position. The stimulating electrode (S) is bent around the freed RLN (arrows) at the level of the 9th tracheal ring. The grounding electrode (G) was sutured into the retracted strap muscles 1 cm away. The arrowhead indicates the crush site.

was stimulated for 10 sec before axotomy in order to determine the minimum voltage threshold for evoking a vocal fold response. The voltage at which vocal cord twitching was elicited was used as the stimulating voltage and ranged between 200 and 300 mV in all cases; the stimulating current was measured to be between 0.1 and 0.2 mA. Immediately after axotomy, the RLN was stimulated proximal to the crush injury site at a frequency of 20 Hz for 30 min. Rat vocal fold movement is different from humans in that there is a constant movement of the folds with inhalation and exhalation. The loss of movements of the left vocal fold was confirmed via endoscopy after every injury. The stimulation parameters were chosen based on previous studies using the rat facial nerve injury model (Hetzler et al., 2008; Lal et al., 2008; Sharma et al., 2009; 2010; Foecking et al., 2012). After the stimulation period, the electrodes were removed and wound sites were closed as described above.

2.3. Hormone administration

Immediately following injury, as previously described, two Silastic capsules (0.062 in. id × 0.095 in. od; 10-mm length), equilibrated in physiological saline and containing 100% crystalline TP (Sigma), were subcutaneously implanted in rats receiving hormone treatment (Hetzler et al., 2008). The dosage given has previously been shown to establish supraphysiological levels of systemic TP (Kujawa et al., 1989; Hetzler et al., 2008; Tanzer & Jones, 2004). Previous studies demonstrated that gonadally intact animals have an average T concentration of about 2.25 ng/mL (Brown et al., 1999); and that rats receiving hormone treatment and produced serum testosterone levels of 10.36 ± 0.32 ng/mL 15 to 19 days post crush injury (Hetzler, 2008).

2.4. Vocal fold mobility assessments

After being anesthetized, animals were secured supine in a stereotaxic skull frame in order to perform the direct laryngoscopy for vocal fold mobility (VFM) measurements. As described by Brown and colleagues, direct laryngoscopy was achieved with a pediatric ENDOGO® and a user-modified sheath (Brown et al., 2012). Videos were taken with the ENDOGO® at the following points during surgery: pre-crush or pre-electrode placement (to confirm normal movement before crush), during determination of threshold ES, immediately post-crush (to confirm lack of mobility), and the day of euthanasia. Because of the tongue suture required for video capture of VFM, we did not perform weekly measurements on the same animals; instead animals were assigned to a time-point and VFM measurements were only performed on the day of the crush and the day of euthanasia.

Recovery of vocal fold mobility on the left side was compared to the intact function on the right side. Videos were blinded and scored by two trained observers who came to a scoring consensus. A scale of 0–4 with half-points was used where 0 represents an immobile left vocal fold 0 or 0% mobility, 1 represents 25% mobility, 2 represents 50% mobility, 3 represents 75% mobility, and 4 represents 100% mobility. Half-points were used when the function was between quartiles. Since some animals had pre-operative VFM scores of 3.5, we decided to consider all animals achieving a score of 3.5 or 4 as completely recovered.

2.5. Statistical analysis

Because every animal in the treated groups recovered by 4 weeks post-operative, only the control group has data from post-operative weeks 5 and 6; therefore, the statistical analysis only included data from the first 4 weeks.

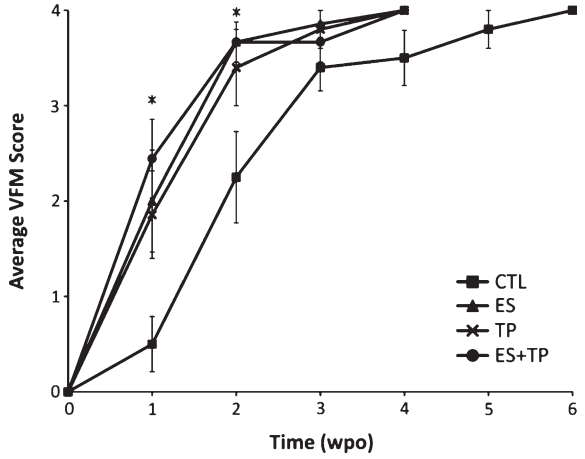


Fig. 2. Effects of brief ES, and/or TP on VFM following a RLN crush injury. Shown are the average VFM scores for 1, 2, 3, and 4 weeks after RLN crush injury comparing all treatments. Each data point represents a separate cohort of animals. Vertical lines represent standard error of the mean. $n = 4$–9/group.

Mean VFM scores were analyzed by two-way ANOVA, followed by *post hoc* tests using the Holm-Sidak method. Treatment group and survival time served as the factors for this analysis (see Fig. 2).

In addition, recovery of vocal fold mobility was examined by determining the proportion of animals from each treatment group and survival time that had achieved complete recovery (VFM score of 4). These data were entered into a three-way contingency table—with treatment group, survival time, and recovery status (recovered or not recovered) serving as the categorical variables. The relationships between these variables were then analyzed utilizing a log-linear χ^2 analysis (see Table 1).

3. Results

3.1. Functional recovery

3.1.1. Mean vocal fold mobility

As expected, VFM scores rose over time as animals recovered function [overall test for the effect of survival time on mean VFM score $F(3, 80) = 35.37$, $p < 0.001$]. Furthermore, recovery of vocal fold mobility benefitted significantly from every treatment [overall test for the effect of treatment group on mean VFM score $F(3, 80) = 6.57$, $p < 0.001$]. Interestingly, at no survival time were VFM scores from any of the treated groups significantly different from any other treated group, suggesting that the treatments each had virtually the same efficacy on recovery of vocal fold mobility. By the first post-operative week, control animals had a mean VFM score of 0.50. At the same survival time, mean VFM scores from every treated group were significantly higher than control (ES-only 2.00; TP-only 1.86; ES + TP 2.44; ps < 0.05). By 2 weeks post-operative, control animals had a mean VFM score of 2.25. At this survival time, mean VFM scores from both the ES-only group and the ES + TP group (both 3.67) were significantly higher than control (ps < 0.05); the mean VFM score in the TP-only group, while very similar to the other treated groups (3.40), did not reach statistical significance in comparison to control. By 3 and 4 weeks post-operative, mean VFM scores were reaching a ceiling in all groups, and there were no significant differences between groups at these time points. Finally, there was no significant interaction between treatment group and survival time [$F(9, 80) = 0.90$, *ns*].

Table 1
Percentage of animals achieving complete functional recovery following a RLN crush

Treatment group	Weeks post operative			
	1	2	3	4
Untreated	0	0	40	50
TP	22	60	83	83
ES	22	83	100	100
ES + TP	33	83	83	100

3.1.2. Proportion at full recovery

Table 1 shows the percentage of animals in each group that had reached full recovery (VFM score = 4) at each survival time. The χ^2 analysis revealed a significant main effect of treatment group [$\chi^2(3) = 9.00$, $p < 0.05$], a significant main effect of survival time [$\chi^2(3) = 37.96$, $p < 0.0001$], and a significant interaction effect [$\chi^2(24) = 59.26$, $p < 0.0001$]. In the control group, no animals reached full recovery in the first 2 weeks post-operative. In contrast, at least 22% of the animals in each of the treated groups had reached full recovery by 1 week, and at least 60% had done so by 2 weeks. By 4 weeks post-operative, only 50% of control animals had reached full recovery (in fact, it took a full 6 weeks for 100% of control animals to reach full recovery; data not shown). In contrast, 83% of TP-only animals, and a full 100% of both ES-only and ES + TP animals had reached full recovery by 4 weeks post-operative.

4. Discussion

The present study found that administration of monotherapy of ES and the combination of ES and TP both had similar effects on functional recovery. ES alone accelerated the VFM functional recovery parameter and was sufficient to improve complete functional recovery as compared to untreated animals. In contrast to previous studies in the facial nerve, the combination of ES and testosterone in the RLN crush injury model, however, did not significantly shorten the time until complete functional recovery more than individual treatment. There are a couple of factors which might explain the difference in treatment results between the facial and recurrent laryngeal nerve injury models. One difference which could account for this is the particular nerve injured. The facial and recurrent laryngeal nerves are similar in that the motor components of these nerves are brainstem in origin. Where they differ is their function. The facial nerve is completely motor once it exits the facial canal and experimentally is usually injured distal to this point. In contrast, both motor and sensory fibers of the RLN travel with the vagus nerve for some distance prior to branching off at the level of the aortic arch on the left and the subclavian artery on the right, finally terminating in the larynx. The polyvagal theory has suggested that the myelinated and unmyelinated portions of the nerve serve different functions for neurogenic control of the heart (Porges, 2009). Since the heart is extremely sensitive to electrical activity, the vagal neurons and fibers that eventually compose the RLN may be more responsive to electrical stimulation as a therapy. This could account for the seeming lack of improvement with the addition of TP to the treatment regimen that is necessary in studies of regeneration of the facial nerve (Sharma et al., 2010a). Additionally, the distance to regenerate in the RLN model is almost twice as short as that required in an extratemporal facial nerve crush (the most distal crush of a cranial nerve our laboratory has examined). If the required regeneration distance is short, to a certain degree maximal acceleration of regeneration may be achieved by one therapy alone and thus no extra benefit is seen by combining two effective treatments.

In a number of different model systems, electrical stimulation of has been demonstrated to accelerate behavioral recovery from facial and hindlimb paralysis after injury to the facial, femoral, and sciatic nerves (Al-Majed et al., 2000; Hetzler et al., 2008; Lal et al., 2008; Gordon et al., 2009; Huang et al., 2010; Franz et al., 2013). Direct stimulation of the injured nerve stump has been shown to decrease the delay in axonal sprouting post-injury as well as upregulate the regeneration-associated genes GAP-43, α_1-tubulin, BDNF, PACAP, and neuritin to enhance regeneration (Sharma et al., 2009, 2010b; Wenjin et al., 2011). Furthermore, this treatment would be easily applied in a surgery setting as a stimulator could be connected to the intraoperative monitors used to detect nerve damage during neck and thoracic operations. If nerve damage is noted, the stimulation could be applied while the patient is still under anesthesia. The disadvantage of a subsequent increase in operative time would be more than compensated for by the potential for improved speech, swallowing, and breathing outcomes with the use of ES therapy. Additionally, though the 30-min duration of stimulation has been shown to be effective in this study using the RLN and in others using the facial nerve, shorter periods have not be examined and it is entirely possible that less time may be required to see positive results in humans.

Gradual recovery in VFM after unilateral RLN crush injury was observed over a period of 6 weeks in untreated animals and 4 weeks in treated animals. The untreated animals recovered along a similar time course to that observed in other studies using a unilateral crush injury (Araki et al., 2005; Tessema et al., 2008, 2009). Tessema et al., noted moderate recovery by 3 wpo (2008) with full recovery by 6 weeks postoperative. Our observation that none of the animals in the 2 wpo time point had recovered and only half of the animals had completely recovered at the 4 wpo time point confirms others reports (Araki et al., 2005; Shiotani et al., 2007).

It is also important to be aware of the relative differences in the distances that RLN motoneurons must regenerate their axons in humans versus rats when interpreting the results of the acceleration of behavioral recovery (both initiation and complete). Both human and rat nerve regeneration rates are in the same order of magnitude, approximately 1.7 mm/day in humans (Sunderland, 1947) and 4 mm/day in rats (Kujawa et al., 1993; Wang et al., 1997); however, the distance that a RLN motoneuron in a rat must regenerate its axon is on the order of millimeters whereas the same injury in a human will require the RLN motoneuron to regenerate centimeters of axon. Given this data, it is possible that humans will respond differently and will require the combination treatment to achieve

more complete recovery because their RLN injuries are comparatively more severe than those delivered to the rats in this pre-clinical study.

Source of financial support

Loyola Research Funding Committee Grant

Acknowledgments

This research was financially supported by Loyola University Research Funding Committee Grant (#201299). We would like to thank the Department of Otolaryngology at Loyola University Medical Center for providing support for this study.

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